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Econometric modeling for the influence of economic variables on secondary copper production in India

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ABSTRACT

The stocks and flows of primary metal resources, and the estimates of so generated secondary scrap metal flows are tracked and analysed utilizing the traditional analytical method of 'material flow analysis'. However, the scrap availability and hence the secondary metal production is also influenced by fluctuating economic conditions. Here we utilize the autoregressive distributed lag approach to model the influence of selected economic variables on secondary metal production and hence the scrap availability in the country. We find electricity index, mining index, and electric power generation and distribution correlate with secondary copper production in long-run; and electricity index correlate with secondary copper production in short-run. A 1% increase in electricity index initially leads to 1.76% drop in secondary copper production in short-run, and 3.69% increase in secondary copper production in long-run. A 1% increase in both mining index, and electric power generation and distribution leads to 4.57% and 5.18% decrease in secondary copper production in long-run respectively. The CUSUM and CUSUMSQ charts show the developed ARDL model to be stable over time. Based on the model dependence, we suggest that the policy makers shall devise policies towards reducing copper mining activities thereby encouraging stakeholders to focus on effective collection and efficient recycling of copper towards conservation of natural metal resources.

1. Introduction

Ambitious targets on economic growth have led to the increased use of electrical and electronic equipments (Mhatre et al., 2021; Panchal et al., 2021) especially in developing nations like India. Copper metal is highly utilized in the manufacturing of these equipments. This high demand coupled with the high nature displacements have resulted in depletion of copper resources in the world. Hence, this metal resource is classified as 'deficit category' due to the concerns on its limited ore reserves (FICCI, 2013; IMY, 2019; Sourabh et al., 2023). The indigenous copper production in India has declined by 40% between 2015 and 2019 (CMIE, 2020). Moreover, copper imports to India has increased at CAGR of 10.65% in the last five years thereby transforming the country to a net importer of copper (CMIE, 2020; MoM, 2020; The Economic Times, 2019). Recently, the occurrence of *limits to growth* kind of behaviour is evidenced in the indigenous copper production in India between the time period 2014 to 2019 (CMIE, 2020). It is expected that India will become the 6th largest copper market by 2030 with electrical, transport and telecommunication sectors emerging as major copper consumers in the country (ICSG, 2021). The copper demand in India is expected to grow further at 6–7% due to Government of India's "make in India" policy, "Smart City" programmes, development of industrial corridors, housing for all Indians by 2022, National Highway development project, rail project, and defense production policy to encourage indigenous manufacture (MoM, 2020). This on-going nature utilization scenario resulting in high depletion rates of metal ore resources has motivated the developing economies like India to emphasize on transformation from "take-make-use-dispose" scenario or "extract-produce-consume-trash" industrial model (Desing et al., 2020; Ellen MacArthur Foundation, 2015; Ghisellini et al., 2016; Merli et al., 2018; Sauvé et al., 2016) to "take-make-recycle" scenario towards meeting the ever-increasing demand. Thus, the above facts motivate the authors to conduct the present research on secondary copper production and recycling Industry in India.

From the environmental and economic dimensions of sustainability (Georgiadis, 2013), collection and recycling of copper scrap is desirable as the scrap is often cheaper than the primary metal as it does not incur any exploration and primary refining cost along with lower energy

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requirements (Agrawal and Sahu, 2010; Blomberg and Söderholm, 2009). Moreover, the advantage of conservation of natural resources has been a further motivation for copper recycling in the country (Agrawal and Sahu, 2010). Hence, in the last decade, the importance and need for recycling has increased the research on understanding various technical (Bonnon et al., 2015; Jamali et al., 2022; Wang et al., 2020), economic (Achillas et al., 2013; Blomberg and Söderholm, 2009; Ghodrat et al., 2016) and life cycle (Jingjing et al., 2019; Pokhrel et al., 2020) aspects of improving recycling around the globe (Fu et al., 2017). Moreover, the research works on recycling are mainly realized through the traditional material flow analysis (MFA) and less frequently through econometric studies. A large pool of recycling research utilizing MFA exists to date. The MFA is used to study recycling efficiency of waste management systems (Bertram et al., 2002) and dissipative releases from metals from an environmental impact perspective (Lifset et al., 2012). These MFA models estimate material recovery by incorporating time series of processing rate, recycling rate, scrap ratio, material consumption and product end-of-life (Bertram et al., 2002; Chen, 2013; Cullen and Allwood, 2013; Gsodam et al., 2014; Habuer et al., 2014). The results of these MFA can aid the government and policy makers to devise strategies towards increasing the recycling rates (Fu et al., 2017; Wang et al., 2014). Hence, the MFA mainly involves assessing the past and present material stock thereby estimating the future material stocks and flows in the anthroposphere (Müller et al., 2014). As such, MFA assumes that the past and present behaviour of material stock will continue to the future (Fu et al., 2017) and do not consider the influence of changing economic variables in the material flow dynamics.

The dynamics of metal markets is researched from the point of view of finance (Baffes and Savescu, 2014; Ghodrat et al., 2016), environmental impact (Bonnin et al., 2015; Jamali et al., 2022; Raghupathy and Chaturvedi, 2013; Rubin et al., 2014; Wang et al., 2020) and resource economics (Slade, 1980; Solow, 1974). The econometric studies on resource recovery and recycling can be traced back to Fisher et al. (1972) and Slade (1980). While the former study attempted to capture the supply and demand side of primary copper supply; the latter focused on econometric modeling for secondary copper recovery involving scrap generation and accumulation. There exist only a few econometric studies on metal recycling from the primary industry perspective (Alonso et al., 2012; Fisher et al., 1972) or the secondary industry perspective (Blomberg and Söderholm, 2009; Slade, 1980). The studies on secondary metal industry have shown that the secondary production correlates with explanatory variables of scrap price (Blomberg and Söderholm, 2009; Tan, 1987), refined metal price (Fu et al., 2017; Gómez et al., 2007; Rivera et al., 2023; Valencia, 2005), energy cost (Vial, 1988), scrap stock (Gómez et al., 2007; Rivera et al., 2023; Valencia, 2005; Vial, 1988), scrap flow (Gómez et al., 2007; Rivera et al., 2023), difficulty in collecting scrap (Rivera et al., 2023; Valencia, 2005), income from secondary activity (Vial, 1988), lagged secondary production (Fu et al., 2017; Gómez et al., 2007; Rivera et al., 2023; Tan, 1987), GDP (Fu et al., 2017) and mining index (Fu et al., 2017; Rivera et al., 2023). The studies aid the governments and policy makers in analysing market behaviour and policy impacts pertaining to secondary metal industry. These studies are also important while designing policies towards improving metal recovery and recycling in the long term. Still the research focusing on the influence of changing economic variables on metal recycling is sparse and is at its infancy specifically with India. Hence this work is novel in the fact that no previous attempt has been made to identify and quantify the effects of certain relevant influential economic variables on the secondary copper production in India, and also explore its long and short-run effects if at all any. Moreover, being one of the initial studies that attempts to model the influence of economic variables on secondary copper production in Indian context, our study opens up a line of inquiry that is very valuable for decision makers. Our model can explain how the changes in the economic conditions influence the secondary copper industry in the country. The existing econometric studies on secondary metal production are found to compromise on the explanation of long-term trends when the models miss short-run changes. This usually occurs when short-run changes of the market may displace the long-term trend of the market which is missed by the existing econometric models (Rivera et al., 2021). For the same reason these econometric models fail to explain long-term trends. The present model explains both the short-term and long-term trends in secondary copper production under the influence of changing economic conditions in the country. In addition, an important attribute for econometric models is its robustness. The developed econometric model is stable over time as evidenced through the stability tests and hence it is a robust model.

2. The data

The data utilized to study the dynamic relationship between the dependent and the independent variables include annual time series data for a period of 38 years from 1983 to 2020. The data was collected from the Indian government and scientific community accepted data banks of World Bank (2020), International Copper Study Group (ICSG, 2021), Reserve Bank of India (RBI, 2020), Indian Bureau of Mines (IBM, 2021), International Monetary Fund (IMF, 2020), Trend Economy (2021), Forex (2020) and FRED (2020).

The dependent variable considered is annual secondary copper production (Q_t) in the country. Previous studies have considered scrap supply, scrap availability and scrap stock (Blomberg and Söderholm, 2009; Fisher et al., 1972; Fu et al., 2017; Glöser et al., 2013; Gómez et al., 2007; Rivera et al., 2021; Slade, 1980; Valencia, 2005; Vial, 1988) as dependent variables that are all proxies for secondary copper production. In this study, the secondary copper production implies copper scrap in both organised and unorganised sectors of the country. Also, copper scrap or secondary copper production (which is the copper produced from secondary sources) is considered as a whole and has not been separated into refined scrap (scrap that is first refined before use) or direct melt scrap (directly melted scrap at the time of use). Based on the report by Indian Bureau of Mines (IBM, 2011), the secondary copper production in India was estimated as 20 percent of primary copper production for the years 1983–2020.

The predictor variables for the secondary copper production are identified from literature on various econometric models pertinent to metal markets (Fisher et al., 1972; Slade, 1980; Tan, 1987; Vial, 1988; Valencia, 2005; Gómez et al., 2007; Fu et al., 2017) developed to capture the secondary refined copper supply/production worldwide. Moreover, as suggested by Fu et al. (2017), the variables are selected with the aim to have a comprehensive coverage of activities that may influence the Indian secondary copper industry. In addition, emphasis has been laid in selecting the appropriate set of independent variables by considering the socio-economic conditions prevailing in the country. As such, twelve explanatory variables considered in the model are gross domestic product (GDP), world copper price for refined copper, exchange rate, manufacturing output, electric power generation and distribution, mining index, manufacturing index, electricity index, import of primary copper, export of primary copper, import of secondary copper and export of secondary copper. The GDP (Y_t) is a measure for economic output of a country (Menon, 2017) and previous studies have established the influence of economic growth rate on recycling patterns (Binder et al., 2006; Fu et al., 2017; Van Beukering and Bouman, 2001). The GDP is expressed in 'billions of USD' from IMF (2020). The world copper price (P_t) signifies the refined copper price reflected on London Metal Exchange (LME). As the copper price in the Indian market is completely dependent on the LME prices, and considering the ease of availability and more authenticity of LME prices, it finds its way as one of the independent variables considered in the study. There also exist econometric models that incorporates world copper price as an explanatory variable (Fisher et al., 1972; Fu et al., 2017; Slade, 1980; Tan, 1987; Vial, 1988). The world copper price is in 'US dollars per metric tons' from FRED (2020). The exchange rate (ER_t) is a key

measure that affects the decisions of investors and other businesses (Dua and Ranjan, 2012). The exchange rate is a crucial factor affecting Foreign Direct Investment (FDI), imports, and exports of various goods (including copper ore and scrap). Also, the role of exchange rate in swaying the inflation rate will influence the purchase of consumer goods such as electrical and electronic appliances, automobiles etc. which have a significant copper consumption. This, in turn, will affect the amount of copper scrap that would be available in the country after their end of lives. Hence, exchange rate is considered to understand the monetary influence in the Indian secondary copper market.

The major share of copper usage in India goes to the Electric and Telecom sector. Electric power generation and distribution, for a booming economy coupled with high population like in India, has been increasing at a substantial rate over the past decades. This growth has resulted in the increased copper usage in terms of high-grade copper cables for transmission and distribution. Presently, more than 34% of copper consumption in India is concentrated in electrical industry (Agrawal and Sahu, 2010). Hence electric power generation and distribution (EG_t) which cater for this vital information is considered as an independent variable. However, the copper utilized in power generation and distribution takes about 40 years before it is recycled as copper scrap (MoM, 2011). Hence this vital variable has been considered in the study to analyse its consequence empirically on secondary copper production. For a developing country like India, copper is heavily utilized in the manufacturing sector in terms of transformers, generators, end products or even auxiliary equipment which assist in manufacturing and also the scrap being generated (ICSG, 2021). This prompted the inclusion of Manufacturing Output (MO_t) as one of the independent variables so as to confirm its influence on secondary copper production in the country. Further it also relates with the copper scrap supply as a whole. Thus, manufacturing output in the present study refers to the net output of a sector after adding up all outputs and subtracting intermediate inputs. Its calculation does not include the deductions for depreciation of fabricated assets or depletion and degradation of natural resources. The same has been included to find out if the manufacturing productivity in the country influences secondary copper production in the long and short terms.

The three contributing indices towards India's Industrial Production (IIP) Index are Electricity Index (EI_t), Mining Index (MI_t) and Manufacturing Index (MFI_t) (RBI, 2020). One of the major copper consumers in India is the electricity sector (Agrawal and Sahu, 2010). Electricity Index (EI_t) signifies the growth in electricity sector of the country. Electricity index is considered in the present study to understand the extent of influence of electricity sector growth on secondary copper production in the country. The Electricity Index data are obtained from Reserve Bank of India (RBI, 2020). Mining Index (MI_t) shows the output of the entire mining industry (Rivera et al., 2021) and hence signifies the growth in mining sector of the country. The mining index data are retrieved from Reserve Bank of India (RBI) reports between 1983 and 2020. It is a composite index for mining computed by Indian Bureau of Mines using 29 to 60 minerals. Mining index is considered in the present study to identify if secondary copper production is considerably influenced by the mining activity in the country. Manufacturing Index (MFI_t) signifies the growth in the manufacturing sector of the country. For obtaining this index, the output value of 809 items in accordance with National Industrial Classification was obtained from Annual Survey of Industries (RBI, 2020). For any model to cater for recovery economic equilibrium of secondary production in the country, it is paramount that the imports and exports also be considered. Hence, import of primary refined copper (IPR_t), export of primary refined copper (EPR_t), import of secondary copper (IS_t) and export of secondary copper (ES_t) are included in the model. The variables considered in the model and its sources are provided in Table 1.

Table 1
Variables in the model.

Variables	Symbol	Unit of Measurement	Source
Secondary copper production	Q	Tons	IBM (2021)
Electricity Index	EI	Dimensionless	RBI (2020)
Manufacturing Index	MFI	Dimensionless	RBI (2020)
Mining Index	MI	Dimensionless	RBI (2020)
Manufacturing output	MO	Tons	World Bank (2020)
Gross domestic product	Y	Billions of USD	IMF (2020)
Exchange rate	ER	Indian Rupee (INR)	Forex (2020)
World copper price	P	US$/mT	FRED (2020)
Electric power generation and distribution	EG	kWh/capita	World Bank (2020)
Primary copper imports	IPR	Tons	IBM (2021)
Primary copper exports	EPR	Tons	IBM (2021)
Secondary copper imports	IS	Tons	Trend Economy (2021)
Secondary copper exports	ES	Tons	Trend Economy (2021)

3. Research methodology and theoretical model

The research methodology is adopted from Fu et al. (2017) and is appropriately adapted for the present research. The same is presented as a flowchart in Fig. 1.

Initially the variables are logarithmically transformed to make the error terms normal (Menon, 2017). Jarque-Bera test is utilized for testing data normality. The stationarity of the variables is checked using Kwiatkowski-Phillips-Schmidt-Shin (KPSS) unit root test (Arltová and Fedorová, 2016).

Following the selection of predictor variables, forward stepwise regression is used to identify the most influential and statistically significant variables. The results of forward stepwise selection are utilized in developing the ARDL model. The ARDL model was utilized as it can be applied when the underlying variables are stationary in 0 or 1 or a combination of both. It is more efficient for small and finite data sizes (Haug, 2002). Moreover, error correction model (ECM), which provides short as well as long run dynamics, is derived from ARDL through simple linear transformation (Banerjee et al., 1993). The next step identified the

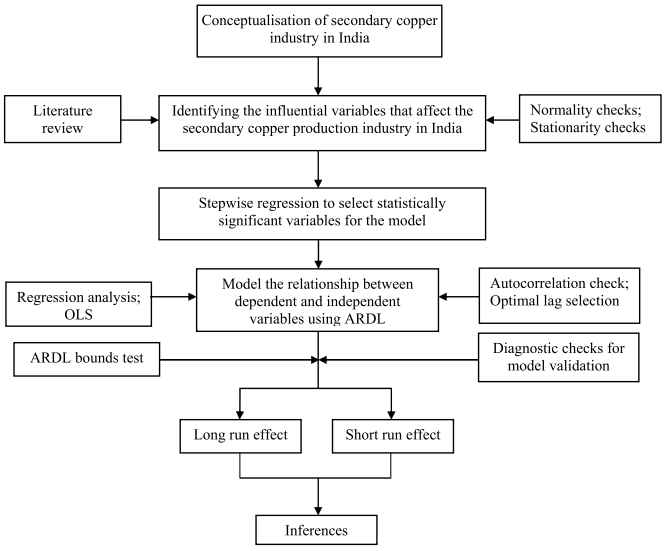


Fig. 1. Research framework.

optimal lag order to be considered in the ARDL model. The lag selection was carried out by comparing a multitude of lag selection criteria such as Akaike Information Criteria (AIC), sequential modified LR test statistic at 5% level, Final Prediction Error (FPE), Schwartz Information Criteria (SIC), and Hannan–Quinn information criteria (HQIC), thereby selecting the one with the lowest value. Cointegration bounds test is performed to obtain the existence of long run and short effects in the model. Finally, diagnostic checks utilizing Breusch Godfrey Lagrange Multiplier (LM) test (Godfrey, 2007), Breusch-Pagan-Godfrey test (Breusch and Pagan, 1979), Ramsey RESET test (Peters, 2000), CUSUM and CUSUMSQ are performed.

4. Estimation and results

The empirical results were obtained using E-Views 10. The descriptive statistics of all the variables are shown in Table 2. The null hypothesis for Jarque-Bera normality test is "residuals are normally distributed" against the negation hypothesis of non-normal residuals (Jarque and Bera, 1980). The Jarque-Bera probability (Table 2) shows that the residuals are normally distributed. The test for stationarity of the variables are depicted using Kwiatkowski-Phillips-Schmidt-Shin (KPSS) unit root test (Table 3). The study variables displayed trend component and hence "intercept with trend" option of unit root test was utilized. The result shows that the LM-statistic (Table 3) for the variables are less than the asymptotic critical value at 1 percent significance level thereby accepting the null hypothesis of series being stationary.

The forward stepwise regression estimation yielded three statistically significant variables out of twelve input variables at a 5% level of significance (Table 4). The selected three variables from the forward stepwise selection output are Electricity Index (EI_t), Mining Index (MI_t) and Electric power generation and distribution (EG_t). These three selected variables are the input exogenous variables in the further developed Autoregressive Distributed Lag (ARDL) model. An ARDL model with the three independent variables is depicted mathematically in Eq (1), where 'l' is the number of optimal lags to be considered in the model and 't' the current time period.

$$Q_t = \alpha + \sum_{i=1}^{l} \beta_i\, Q_{t-i} + \sum_{i=0}^{l} \delta_i\, EI_{t-i} + \sum_{i=0}^{l} \mu_i\, MI_{t-i} + \sum_{i=0}^{l} \theta_i\, EG_{t-i} + \varepsilon_t \tag{1}$$

As the predictor and response data displayed linear trend component, the econometric analysis may not efficiently capture the causal dynamics between the variables resulting in spurious regressions (Fu et al., 2017). Hence de-trending was effected by regressing the 'time' (T) component as an independent variable in the ARDL model (Eq. (2)).

$$Q_t = \alpha + \sum_{i=1}^{l} \beta_i\, Q_{t-i} + \sum_{i=0}^{l} \delta_i\, EI_{t-i} + \sum_{i=0}^{l} \mu_i\, MI_{t-i} + \sum_{i=0}^{l} \theta_i\, EG_{t-i} + \sum_{i=0}^{l} \varphi_i\, T_{t-i} + \varepsilon_t \tag{2}$$

The forward stepwise regression estimation resulted in Durbin-Watson statistic of 0.538 (Table 4) indicating the presence of positive autocorrelation. In order to remove autocorrelation, variables were lagged after identifying the appropriate lag order utilizing the different methods detailed in section 3. The lag order selection criteria involve selecting the one with the lowest value. The optimal lag 2 has least values of AIC, SIC and HQIC as in Table 5. Moreover, AIC value corresponding to 2 lags was found to be the lowest and hence, 2 lags is considered to be optimal for the ARDL model. Once the lag order corresponding to 2 lags using AIC was selected (based on the lowest AIC score −0.253), ARDL model (Eq (2)) is revised to incorporate two lags for dependent and independent variables respectively as in Eq (3).

$$\begin{aligned} Q_t &= \alpha + \sum_{i=1}^{2} \beta_i\, Q_{t-i} + \sum_{i=0}^{2} \delta_i\, EI_{t-i} + \sum_{i=0}^{2} \mu_i\, MI_{t-i} + \sum_{i=0}^{2} \theta_i\, EG_{t-i} + \sum_{i=0}^{2} \varphi_i\, T_{t-i} + \varepsilon_t \\ Q_t &= \alpha + \beta_1\, Q_{t-1} + \beta_2\, Q_{t-2} + \delta_0\, EI_t + \delta_1\, EI_{t-1} + \delta_2\, EI_{t-2} + \mu_0\, MI_t + \mu_1\, MI_{t-1} + \\ &\mu_2\, MI_{t-2} + \theta_0\, EG_t + \theta_1\, EG_{t-1} + \theta_2\, EG_{t-2} + \varphi_0 T_t + \varphi_1 T_{t-1} + \varphi_2 T_{t-2} + \varepsilon_t \end{aligned} \tag{3}$$

The ARDL model with lag order (2,2,0,1,1) is selected out of 162 models. The result is depicted in Table 6 with its criteria graph (Fig. 2) which show the top 20 models selected based on lowest AIC score. The variables secondary copper production, electricity index, mining index, electric power generation and distribution, and time have a lag of order 2,2,0,1 and 1 respectively. The dependent variable secondary copper production is significant with two lags having probability value less than 0.05 (Table 6). Hence, the model in Eq (3) is further revised by incorporating the obtained variables with suitable lags that forms the best model. Eq. (4) gives the final ARDL model to be estimated.

$$\begin{aligned} Q_t &= \alpha + \beta_1\, Q_{t-1} + \beta_2\, Q_{t-2} + \delta_0\, EI_t + \delta_1\, EI_{t-1} + \delta_2\, EI_{t-2} + \mu_0\, MI_t + \theta_0\, EG_t + \\ &\theta_1\, EG_{t-1} + \varphi_0 T_t + \varphi_1 T_{t-1} + \varepsilon_t \end{aligned} \tag{4}$$

The adjusted R^2 value of the estimated ARDL model (Eq (4)) is 0.987 (Table 6) which signifies that the selected three independent variables explain maximum percentage of variation with the dependent variable. The autocorrelation evidenced prior is removed by the addition of sufficient lags in the model. The same is validated from the Durbin-Watson statistic value of 2.245. A value close to two indicates no autocorrelation.

The Cointegration bounds test is performed to analyse the long run and short effects in the model. The F-statistics value (7.314) shown in Table 7 is higher than the upper and lower bound values at all significance level (10%, 5%, 2.5% and 1%) under constant and no trend. Hence, the null hypothesis of no cointegration among the variables is rejected (Aftab et al., 2021) thereby showing evidence of both long-run and short-run effects. As the existence of cointegration is established, the result for estimation of log-linear long-run secondary copper production ARDL (2,2,0,1,1) model is shown in Table 8. All the three predictor variables are statistically significant with probability values less than 0.05. The coefficient values of the variables show the effect of predictor variables on secondary copper production in the context of Indian copper industry for long-run.

The result for short-run impact is shown in Table 9, with the coefficient of cointegration equation statistically significant. This concludes that the variables are statistically significant at 5% significance level (Narayan and Smyth, 2008). The variables of secondary copper

Table 2
Descriptive Statistics of variables.

Variable	Q_t	EI_t	MFI_t	MI_t	MO_t	Y_t	ER_t	P_t	EG_t	IPR_t	EPR_t	IS_t	ES_t
Mean	10.423	4.105	3.949	4.219	4.700	6.547	40.472	8.079	6.086	10.484	12.248	6.228	3.003
Median	11.091	3.982	3.674	4.232	4.348	6.232	43.805	7.921	5.997	10.530	12.436	6.413	3.159
Maximum	11.983	5.065	4.879	4.697	5.994	7.962	74.130	9.086	6.998	11.931	12.844	7.068	4.114
Minimum	7.746	3.227	3.016	3.478	3.568	5.357	10.100	7.225	5.110	9.680	10.517	3.546	1.572
Std. Dev.	1.407	0.620	0.581	0.362	0.851	0.878	18.631	0.628	0.524	0.672	0.692	0.885	0.688
Skewness	−0.417	0.088	0.356	−0.415	0.248	0.288	−0.0510	0.178	0.026	1.025	−1.681	−1.842	−0.411
Kurtosis	1.556	1.606	1.730	1.962	1.469	1.556	2.1008	1.462	2.064	3.019	4.540	5.719	2.293
Jarque-Bera	4.406	3.125	3.356	2.796	4.101	3.830	1.3029	3.948	1.391	6.651	21.639	33.204	1.862
Probability	0.110	0.209	0.187	0.247	0.127	0.147	0.5213	0.139	0.499	0.035	0.000	0.000	0.394

Table 3
KPSS unit root test results for stationarity.

Variable	Q_t	EI_t	MFI_t	MI_t	MO_t	Y_t	ER_t	P_t	EG_t	IPR_t	ERP_t	IS_t	ES_t
LM-statistic[a]	0.160	0.106	0.118	0.197	0.140	0.148	0.162	0.0865	0.096	0.160	0.104	0.102	0.088

[a] *Note*: The critical value (intercept with trend) at 1% is 0.216 (Kwiatkowski et al., 1992).

Table 4
Forward stepwise selection results.

Variable	Coefficient	Standard Error	t-Statistics	Probability
Electricity Index	1.529	0.245	6.233	0.000
Mining Index	4.317	0.802	5.382	0.000
Electric power generation and distribution	−2.311	0.573	−4.035	0.000
Adjusted $R^2 = 0.909$ Durbin-Watson = 0.538 α = 5%				

Dependent Variable: Secondary copper production.
Method: Stepwise Regression.
Included Observation: 38.
Number of searches regressors: 12.
Selection method: Stepwise forwards.
Stopping criteria: p-value forward/backward = 0.05/0.05.

Table 5
Lag selection for significant variables.

Time lag	LogL	LR	FPE	AIC	SIC	HQIC
0	−18.532	NA	0.207	1.266	1.401	1.312
1	4.484	40.618	0.056	−0.028	0.151	0.032
2	9.299	8.213[a]	0.045[a]	−0.253[a]	−0.028[a]	−0.176[a]
3	9.695	0.035	0.048	−0.1.95	0.074	−0.103

[a] Indicates lag order selected by criterion.

Table 6
ARDL model result.

Variable	Coefficient	Standard Error	t-Statistics	Probability
Secondary copper production (−1)	1.029	0.136	7.548	0.000
Secondary copper production (−2)	−0.457	0.134	−3.406	0.002
Electricity index	0.686	0.593	1.157	0.258
Electricity index (−1)	−0.867	0.819	−1.058	0.300
Electricity index (−2)	1.760	0.578	3.047	0.005
Mining index	−1.958	0.786	−2.490	0.020
Electric power generation and distribution	1.970	1.756	1.122	0.272
Electric power generation and distribution (−1)	−4.185	1.823	−2.288	0.031
Time	6.391	2.597	2.461	0.021
Time (−1)	−3.998	2.022	−1.978	0.060
C	12.701	2.756	4.610	0.000
Adj $R^2 = 0.987$ Durbin-Watson statistics = 2.245 F- statistics = 280.234 (0.000)				

*Note: *p*-value and any subsequent tests do not account for model selection.
Dependent Variable: Secondary copper production.
Method: ARDL.
Included observations: 38.
Maximum dependent lag: 2 (Automatic selection).
Model selection method: Akaike info criteria (AIC).
Dynamic regressors (2 lags, automatic): Electricity index, Mining index, Electric power generation and distribution, Time.
Number of models evaluated: 162.
Selected Model: ARDL (2,2,0,1,1).

production with 1 lag and electricity index with 1 lag have probability values less than 0.05 and hence are statistically significant. The coefficient of cointegration value implies that secondary copper production of

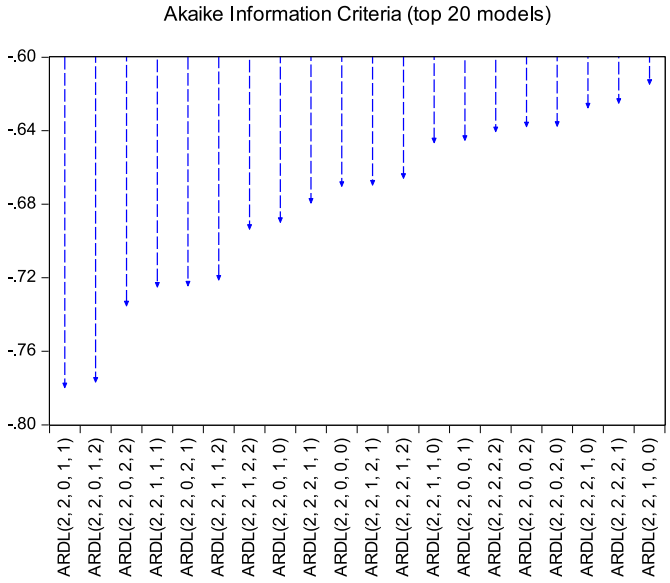


Fig. 2. Criteria graph for the selection of ARDL model.

Table 7
ARDL cointegration test.

F-Bounds Test	Null Hypothesis: No level relationship			
Test Statistics	Value	Significant	I (0)	I (1)
Asymptotic: n = 1000				
F-statistics	7.314	10%	2.45	3.52
K	4	5%	2.86	4.01
		2.5%	3.25	4.49
		1%	3.74	5.06

Table 8
Long-run estimates using ARDL model.

Level Equation Case: Unrestricted Constant and no trend				
Variable	Coefficient	Standard Error	t-Statistics	Probability
Electricity index	3.691	0.997	3.702	0.001*
Mining index	−4.573	1.961	−2.332	0.028*
Electric Power generation and distribution	−5.175	0.929	−5.569	0.000*
Time	5.589	1.424	3.925	0.001*
C	12.701	2.756	4.609	0.000*

Note: ARDL model 2,2,0,1,1, * *p*-value significant at 5%.

current period is corrected by 42.8% in next period to restore equilibrium.

The estimated ARDL model is tested for serial correlation, heteroscedasticity and stability to ensure model's fitness. Diagnostic and stability test results are summarized in Table 10, Figs. 3 and 4. The Breusch Godfrey Lagrange Multiplier (LM) test value shows no autocorrelation; and Breusch-Pagan-Godfrey test value gives evidence for homoscedastic model variables. The Ramsey RESET test value shows that the model has no omitted variables and is correctly specified. The CUSUM and

Table 9
Short-run estimates using ARDL model.

ARDL Error Correction Regression (ECM) Dependent Variable: Secondary copper production Selected Model: ARDL (2,2,0,1,1) Case: Unrestricted Constant and No Trend				
Variable	Coefficient	Standard Error	t-Statistics	Probability
C	12.701	1.934	6.569	0.000*
Secondary copper production (−1)	0.457	0.111	4.149	0.000*
Electricity index	0.686	0.434	1.580	0.127
Electricity index (−1)	−1.761	0.463	−3.803	0.001*
Electric power generation and distribution	1.970	1.287	1.531	0.138
Time	6.391	0.976	6.551	0.000*
Cointegration equation	−0.428	0.066	−6.514	0.000*
Adj R^2 = 0.645 Durbin-Watson Statistics = 2.245 F-statistics = 11.610 (0.000)				

*p-value significant at 5%.

Table 10
Diagnostic and stability test results.

Diagnostic Statistics Test	F-Statistics (p-Value)	Inference
Breusch Godfrey LM	0.672	No Serial Correlations
Breusch-Pagan-Godfrey	0.827	No Heteroscedasticity
Ramsey RESET Test	0.136	Model specified correctly
CUSUM	–	Stable
CUSUMSQ	–	Stable

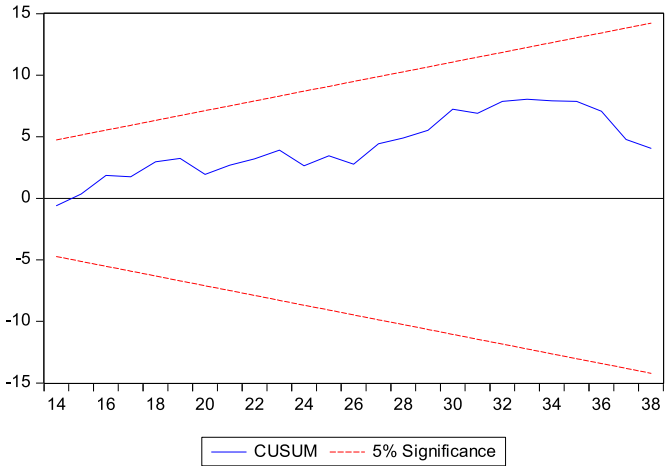


Fig. 3. Plot of cumulative sum of recursive residuals.

CUSUMSQ statistics test are used to check the stability of the coefficients obtained from the long and short run model. The CUSUM and CUSUMSQ plots in Figs. 3 and 4 respectively lie within the critical bands of 5% significance level thereby signifying that the coefficients of the model are stable.

5. Discussions

This paper has empirically examined the impact of electricity index, mining index and electric power generation and distribution on secondary copper production in the context of India by utilizing the concept of ARDL for time series data from 1983 to 2020. From the econometric analysis it is found that the mining index is having a long-run negative impact on secondary copper production in the country (Table 8). This is empirically evidenced as mining index reflects mining activity in the country and constitutes supply side of primary copper production. Hence an increase in mining index will lead to increase in the availability of primary copper thereby meeting the majority of total copper

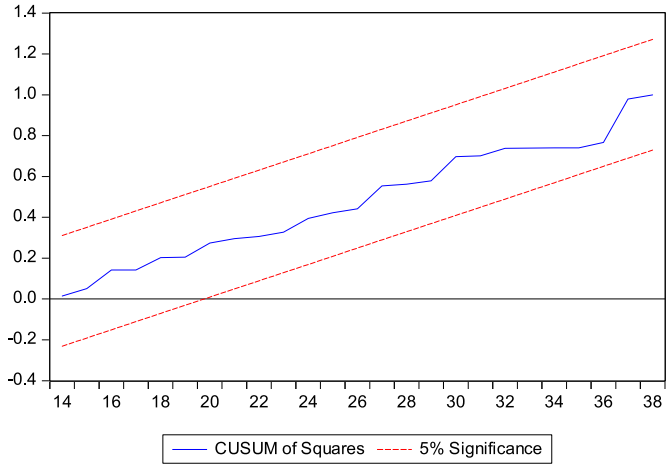


Fig. 4. Plot of cumulative sum of square of recursive residuals.

demand in the country. The so increased availability of primary copper in the market has resulted in the reduced reliance on secondary copper in the Indian economy. This in turn leads to decreased demand for secondary copper. Empirically, 1 percent increase in mining index will result in 4.57 percent decrease in secondary copper production. This in turn shows that the increasing mining index and hence 'nature displacements' are in effect acting as inhibitors to the growth of secondary copper industry in India in the long-run. As empirically evidenced, if the on-going nature displacement scenario continues, it can further result in generating a negative feedback system that will in turn reduce the mining activity due to depleting primary copper resources thereby forcing the system to opt the path of recycling in long-run. To prevent the occurrence of this severe nature depletions in the long-run, more emphasis is to be given towards efficient recycling processes coupled with policies like reduced import duty of metallic scrap which can be processed and/or recycled to ensure higher secondary copper production and utilization in the country.

The electricity index which is a major component of India's Industrial Production Index is empirically found to have a significant long-run positive influence on secondary copper production (Table 8). Empirically, 1 percent increase in electricity index will increase secondary copper production by 3.69 percent in long-run. As the electricity index shows the growth rate of electricity industry, the increase in this index rate will positively influence the secondary copper availability in the economy. Moreover, it also implies that with greater impetus to the electric/power sector growth, the availability of secondary copper scrap can be increased in the country. Thus, policies like tariff controls favouring the industries in power sector coupled with impetus to foreign direct investment in the said sector is bound to have a positive impact on secondary copper production.

Electric power generation and distribution is found to be statistically significant with log-run negative influence on Indian secondary copper production industry (Table 8). Empirically, 1 percent rise in electric power generation and distribution will decrease secondary copper production by 5.175 percent in long-run. Perhaps the trend analysis of dataset shows consistent increase in electricity generation and distribution over the period of this study. This trend infers that with an increase in electricity generation and distribution, the percentage of copper used in electric wires and cables for distribution and domestic use will also significantly increase. However, IBM (2011, page 58) observes that the lead time for copper to contribute towards secondary copper production is very high and may well between 10 and 40 years (including its useful life period) depending upon the type of product (electric cable conductors, heavy sized generators etc). Thus, despite increased demand in copper usage, the metal is available for use in its respective secondary industry (post recycling) only after four decades.

This will negatively impact the availability of secondary copper in the long-run and hence it justifies the negative effect of electric power generation and distribution on secondary copper production industry. This indicates the presence of the so called "hibernating stock" (Daigo et al., 2015) in the copper recycling industry in India. Towards reducing the hibernating scrap, government has to promote addition of scrap as and when it is generated to the secondary production process without much delay through incentives like reduced scrap collection cost and subsidies for scrap collection units. It is also noteworthy that about 3/4th of India's power generation are from conventional sources such as fossil fuels. Thus, to satiate the increasing power requirements of India, government involvement towards cleaner and greener alternatives like solar and hydel power is imperative. Only with such measures will the country be able to sustain the upcoming energy requirements and ensure seamless electricity generation and distribution. This in turn is vital for the red metal's sustained addition to the secondary copper availability in the country after its high lead time.

In the short-run, the first lag of secondary copper production is found to be statistically significant with a positive impact on the response variable. Hence, to improve India's secondary copper production in the short-run, government would need to make more copper scrap available in the market. This may be achieved by reducing taxes on scrap metals being imported. The first lag of electricity index is having significant negative influence on secondary copper production in the short-run. The short-run result suggests that a 1 percent rise in electricity index would result in 1.76 percent drop in secondary copper production. This could be because, for secondary copper production to take place, there should be copper scrap as the raw material. Now, with an increasing growth rate of electricity sector, demand for copper (both primary and secondary) will rise, the copper scrap imports as well as the copper scrap available locally in the country will be hoarded by these companies to avoid disruption in its raw material supply chain. This scrap inventory pile up would leech off the raw material for secondary copper production and hence will not be available for secondary copper production in the short-run as empirically evidenced. It is pertinent to take note that growth in the electricity sector is primarily due to high demand in a fast developing nation like India. Also, as of now the electricity sector of the country primarily thrives on fossil energy. Hence, to ensure that the growth in electricity sector does not inversely affect secondary copper production, government should give more impetus towards minimising transmission and distribution losses. This can be achieved by renewing/replacing various transmission cables and poor associated auxiliaries. This would allow the inherent copper in these inefficient sub systems to circle back into the economy as secondary copper, thus having a two-fold effect in the short-run. Firstly, the secondary copper production in the country would increase as more raw material from the discarded as well as inefficient transmission products will be now available for secondary production. Secondly, the replaced transmission cables and associated auxiliaries will be more efficient. This will reduce electricity losses, which would result in lower power consumption at same or even increased power demands of the country.

6. Conclusion and future research

The present study is based on the secondary copper production in the Indian context and add values to the metal recycling and circular economy literature pool. The present study has contributed with the following.

1. One of the initial studies that attempt to model the influence of economic variables on the secondary copper production in circular economy literature and; the present study is the first of its kind in Indian context.
2. The study explores the long and short-run effects of economic variables on the secondary copper production in Indian context.
3. The study aid in understanding the levers for policy design towards improving copper recycling.

In order to prevent the occurrence of this severe nature depletions and have positive impact on secondary copper production in the long-run, more emphasis is to be given towards efficient recycling processes coupled with policies like reduced import duty of metallic scrap, tariff controls favouring the industries in power sector coupled with impetus to foreign direct investment in the said sector which can be processed and/or recycled to ensure higher secondary copper production and utilization in the country. Further, schemes such as Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY, 2022) through which the electrification of entire rural India is being realized, will act as a catalyst in increasing the growth of electric sector in the country. This in turn would have a positive impact on the secondary copper production in India. There is a huge mismatch between the India's processing requirement and mining capacity and due to this dependence on import of copper has been notified in India. The government should take action to reduce import duty and tariffs on the metallic scrap. It in turn, would have a direct impact on the secondary market industry and spur the growth of entrepreneurs specifically in the field of metal recycling. However, it is imperative that the government also continues to impose stringent environmental regulations in order to obviate the ill effects on the environment by inefficient recycling procedures.

The limitations of this study need to be addressed for future research. The present study was based on copper metal recycling and further similar studies needs to be initiated for other metals to understand whether the outcomes of these studies can be generalized for any metal recycling industry. This study can be extended and improvised to study the impact of price and other potential drivers and suggest policies which would significantly impact copper scrap availability in the country. Further the present economic variables model can be combined with biophysical variables (nature displacement, nature depletion etc.) to develop an integrated biophysical-economic variables model that will capture the interaction dynamics of these variables and the influence these interactions have on metal recycling industry in the country.

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Compliance with ethical standards

The present research does not involve or utilize any Human Participants and/or Animals.

Informed consent

The present research does not involve or utilize any Human Participants and/or Animals.

Code availability

Software codes will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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